VIKTORIIA MAKOVSKA

Domains: NLP, Machine Unlearning, Responsible AI and Data Science

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Summary: Research engineer with a background in software engineering and data science, currently focused on machine unlearning, mechanistic interpretability, representation persistence, and governance of model memory. Former lead software engineer with 7+ years of industry experience, including production software delivery, team leadership, performance optimization, and product collaboration. Current applied research work includes responsible AI, mechanistic interpretability, NLP/LLM evaluation, propaganda mitigation and machine unlearning in LLMs.

TECHNICAL SKILLS

- **AI/NLP systems prototyping:** LLM workflows, RAG, vector search, OpenAI Agents SDK, DeepEval, prompt/evaluation pipelines.
- **NLP and ML engineering:** PyTorch, scikit-learn, FastAI, SFT, AdaLoRA/LoRA/adapters.
- **Responsible AI and governance:** machine unlearning, residual-risk evaluation, participatory data/system design, fairness and interpretability.
- **Backend/data systems:** Python, Ruby, JavaScript, PostgreSQL, Elasticsearch, Redis, FAISS, Pinecone, and Milvus.
- **Cloud and research tooling:** AWS S3/EC2/RDS/ElastiCache, Docker, Weights & Biases, Slurm/sbatch, and multi-GPU experiments.
- **Software delivery:** production stability and delivery management, performance optimization, QA and QC, agile delivery, and stakeholder collaboration.

RESEARCH AND PROFESSIONAL EXPERIENCE

Research Engineer | **Lapa LLM** | *Sep 2025 - Present* | *Open Source* | *https://github.com/lapa-llm/lapa-llm*

- Contributed to Lapa LLM, an open-source project aimed at developing a Ukrainian large language model, with work on machine unlearning for propaganda mitigation and model personality residual trace removal.
- Experimented with intrinsic unlearning approaches including task-vector negation, adapter-based interventions, and neuron-importance-based masking.
- Evaluated trade-offs between intervention effectiveness, residual leakage, representation stability, and model behavior after unlearning.

Research Fellow | **Center for Responsible AI at NYU** | *Sep 2024 - Present* | *Seasonal*

- Conducted research on participatory design of entity-relationship models to mitigate bias in data-driven systems.
- Developed and tested ONION, a participatory ER-modeling methodology that translates stakeholder input into structured system/data models.

Lead Software Engineer | **Artur'In** | *Nov 2019 - Jan 2024*

- Led teams of 10-12 through full software development cycles and delivery of production updates.
- Developed high-performance features for a platform for automatic social-content generation and scheduling aligned with client verticals and preferences.
- Mentored junior engineers and created structured growth roadmaps for QA, backend, and full-stack engineers.

QA, Software Engineer | **Artur'In** | *Sep 2016 - Nov 2019*
- Optimized database queries and resolved performance issues to improve platform speed and reliability (PostgreSQL).
- Supported production stability, reduced downtime, collaborated with product owners on feature priorities, and maintained technical documentation (JQuery, ViewJS, Ruby).
- Contributed to QA and release practices to maintain product stability in an agile delivery environment.

TEACHING EXPERIENCE

Visiting Lecturer | **American University Kyiv** | *Feb 2025 - Present* | *Seasonal*
- Teaching Decision Support Systems and designing practical learning activities on AI-enabled decision support and management-facing analytics.

Course Instructor | **Ukrainian Catholic University** | *Sep 2025 - Present* | *Seasonal*
- Co-design and teaching a Natural Language Processing course, including practical materials on context engineering, RAG, LoRA, and applied NLP workflows.

EDUCATION

PhD Student, Ukrainian Catholic University (UCU) - Ongoing thesis: Forgetting Machines: Responsibility by Design in Machine Unlearning.

Master of Science in Data Science, Ukrainian Catholic University (UCU) - Thesis: Vandalism or Knowledge Manipulation? Detecting Narratives in Wikipedia Edits.

Master in Radio Engineering, Kharkiv National University of Radio Electronics - Thesis: Engine failure detection based on acoustic signal analysis.

LANGUAGES

Ukrainian - Native, English - C1